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Synthesis of C_{5+} hydrocarbons from low H_2/CO ratio syngas over silica supported bimetallic Fe-Co catalyst

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ABSTRACT

In the present study, selective synthesis of C_{5+} hydrocarbons from a low hydrogen to carbon mono oxide ratio syngas has been studied over a novel Fe-Co bimetallic catalysts. Fe was added into Co catalyst to increase the H_2/CO ratio internally via increased water gas shift activity. Catalyst with varying Fe/Co composition (Fe/Co (w/w) = 0.25, 0.5, 1) and varying metal loadings were prepared in the laboratory. Catalysts activity were tested in a fixed bed reactor at T–220 °C, P–20 bar, H_2/CO–1.48 and GHSV–1200 mL/h-gcat. Detail characterization of the catalysts were done through different techniques. Quantitative and qualitative analysis of the liquid products were done using GCMS and CHNS analyzers. The presence of Fe in bimetallic catalyst resulted an increase in CO conversion and product selectivity in comparison to that of mono metallic Fe/Co catalysts. Formation of Fe-Co bimetallic phase was confirmed through the detail characterization of fresh and spent catalysts which facilitate the CO adsorption on active metal sites. However, phases for individual iron and cobalt were also evident at higher metal loadings. The CO conversion increases with metal loading and was maximum at 30% metal loading. Optimum ratio of Fe/Co was 0.5 at a total metal loading of 30% (Cat-2) in which maximum (65%) CO conversion was achieved. Enhancement of H_2/CO ratio through WGS reaction was examined over Cat-2 at various temperature and inlet H_2/CO ratio. At 240 °C, C_{5+} selectivity was maximum at the expense of CO_2 and methane production.

1. Introduction

Conversion of coal and natural gas derived syngas into fuel and value added chemicals via Fischer Tropsch synthesis is being accepted widely. In the past few years, the process has been investigated to develop a catalyst system which makes it selective towards desirable products such as gasoline and diesel range hydrocarbons [1,2]. Syngas which is a combination of H_2 and CO can be obtained from natural gas or gasification of biomass and coal. Biomass and coal derived syngas contains lower H_2/CO ratio as compared to syngas produced from natural gas [3]. For the full utilization of syngas with the lower H_2/CO ratio, it is necessary to enhance the water gas shift activity within the reactor with the help of some modification in the catalyst system.

Eqs. (1) and (2) shows the Fischer Tropsch synthesis (FTS) and water gas shift (WGS) reaction respectively. WGS reaction occurs simultaneously with the FTS reaction, which helps in increasing the ratio of H_2 to CO.

$$nCO + 2nH_2 \rightarrow (-CH_2-)_n + nH_2O \quad (1)$$

$$CO + H_2O \leftrightarrow H_2 + CO_2 \quad (2)$$

The importance of the WGS reaction in Fischer Tropsch synthesis process automatically increased when feed gas contains low H_2/CO ratio (0.5–1.5) because, the overall CO conversion is not very high without the WGS reaction [4].

The most common metal used in Fischer Tropsch reaction are Fe, Co and Ru. Metals such as Ni and Rh also show high activity towards the Reactions (1) and (2). Among these, Ru shows high activity for CO hydrogenation at comparatively lower temperature (<150 °C). However, high cost of Ru limits its industrial scale application. Both iron and cobalt have been used in industry as active FT catalysts since long time due to their economically acceptable cost. These catalysts are being used separately as they possess specific properties to narrow down the product range in a desired hydrocarbon range. Cobalt based catalyst shows high activity and selectivity towards linear long chain hydrocarbons, however, is more expensive than iron. On the other hand Fe based catalyst is mainly selective towards alkene, oxygenates and olefin range

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hydrocarbons which are good petrochemical feedstock. The main property of Fe based catalyst is its high activity towards water gas shift activity which widens its application with wide range of H_2/CO ratio of syngas [5]. This is useful for the utilization of the coal and biomass derived syngas which contain low H_2 to CO ratio. In case of Co catalyst, due to its low water gas shift activity its use is restricted for a specific range of H_2/CO ratio. Therefore, modification in the catalyst system is required to get over with these problems.

It has been shown that by mixing two metal catalysts, the activity and selectivity can be changed drastically [6–11]. The performance of catalyst in the terms of WGS activity and FTS activity was compared with their iron or cobalt based monometallic catalyst. Arcuri et al. [8] presented the results for low CO conversion level obtained with a Fe: Co/SiO_2 catalyst in comparison with the supported mono metal catalyst (T–250 °C and P –1 to14.18 bar). They reported that the dependence of selectivity on conversion does not exist at higher pressure and also iron content in the catalyst resulted methanol as main product at the expense of major methane production. If the CO conversion was taken into consideration the performance of bimetallic catalyst resembles supported iron catalyst at atmospheric pressure level and at 14 bar pressure it matches the supported cobalt catalyst. On the contrary, Ishihara et al. [2] found that the silica based Co-Fe bimetallic catalyst with metal ratio 3:1 shows high selectivity towards C_{6+} hydrocarbons at 250 °C temperature and 10 bar total pressure. They reported that after alloying metal catalyst the activity of the catalyst system enhances due to change in electronic properties of metal species. After detail investigation with the silica supported Co-Ni and Ni-Fe bimetallic catalyst systems Arai et al. [7] reported that, in the FT process Co-Ni bimetallic catalyst increased CO conversion fivefold when compared the same over the single metal catalysts. The use of mesoporous support such as HMS and SBA-15 for the bimetallic Fe-Co catalyst does not show a huge difference in their performances in comparison with the monometallic catalyst and the conversion level ranges between 20 and 30% [12]. It was observed that the Fe/Co bimetallic showed different physicochemical properties than the monometallic Co and Fe catalysts. Addition of Fe increases the reducibility of Co and also increases the number of active sites which in turn affect the CO conversion positively, however, hydrocarbon selectivity did not change significantly. Tihay et al. [13,14] observed very different results. They claimed that during the preparation of the bimetallic catalysts only two phase are formed, Fe-Co bimetallic phase and $CoFe_2O_4$ spinel which are solely responsible for catalytic activity. They reported 5% CO conversion and 84.8% C_2-C_4 selectivity at 10 bar pressure and 220 °C temperature when the catalyst were treated with CO/H_2. Some researchers have shown the effect of water on the FT process and found that the addition of water in feed may help in increasing the WGS activity of catalyst but makes the catalyst system highly prone to deactivation [3,15]. However, all these reports lack detailed investigation of results on FT synthesis and water gas shift reaction over bimetallic catalysts.

The purpose of the present work is to examine the effect of addition of Fe in Co catalyst on the same support and to study their effect in the WGS and FT reactions using CO rich syngas (H_2/CO ratio of 1.0–1.5). The combination of Fe-Co catalyst optimized in terms of metal ratio and metal loading. Identification of different individual metallic phases, bimetallic phases and their impact on the performance of catalyst.

2. Experimental

2.1. Catalysts preparation

Seven different catalysts were prepared via combination of co-precipitation and sol-gel method [16]. First batch of catalysts were prepared with different Fe/Co ratio keeping total metal loading constant at 30%. Then the metal loading was varied keeping the ratio of Fe/Co constant at 0.5. Monometallic silica supported cobalt and iron catalysts were also prepared with the same method. Cobalt nitrate hexahydrate [$Co(NO_3)_2 \cdot 6H_2O$] supplied by MERCK and Ferric nitrate nonahydrate [$Fe(NO_3)_3 \cdot 9H_2O$] provided by Fischer Scientific were used as cobalt and iron precursors respectively. In a typical procedure tetraethyl orthosilicate (TEOS) solution was used as silica precursor. Aqueous ammonia solution of concentration of 25% and nitric acid of 70% concentration were used for the precipitation. Metal catalysts precursor were co-precipitated from the aqueous solution of their nitrate salts and mixed at the desired Fe/Co ratio. Both the salt solutions mixed together and heated at 70 °C for one hour. Liquid ammonia solution was added drop wise to maintain the pH 8 ± 0.2. The precipitate of $Fe(OH)_3$ and $Co(OH)_2$ was filtered and washed repeatedly using deionized water to make its pH neutral. The slurry of the filtered precipitate made in water at 80 ± 2 °C. pH of the slurry adjusted at 9.0 ± 0.5 by adding ammonia solution. Further, desired quantity of TEOS was mixed with ethanol and water in a ratio of 1:4:5 (TEOS: EtOH: H_2O) and stirred for 1 h. Then the solution was added gradually in the previous metal precipitate solution under stirring condition and left for one hour. In the next step, nitric acid was added slowly until the pH reached at 6 ± 0.5. Finally, slurry was filtered, washed and dried over night at 110 ± 3 °C. The dried catalyst samples were calcined at 500 ± 2 °C for 5 h. Similar methods of preparation were applied for all catalysts (i) 5%Fe-25%Co/SiO_2 (ii) 10%Fe-20%Co/SiO_2 (iii) 15%Fe-15%Co/SiO_2 (iv) 30%Co/SiO_2 (v) 30%Fe/SiO_2 (vi) 6%Fe-12%Co/SiO_2 (vii) 3%Fe-6%Co/SiO_2 and these are designated as Cat-1, Cat-2, Cat-3, Cat-4, Cat-5 Cat-6 and Cat-7 respectively.

2.2. Catalyst characterization

Micromeritics ASAP 2010 apparatus was used for determining BET surface area. Pore volumes, average pore diameter of the catalysts were also measured in the same apparatus. Prior to gas adsorption measurements, degassing of the sample was done in which catalyst was heated at 150 ± 1 °C under vacuum for a period of 2 h [17]. The final elemental composition of the catalyst was determined using EDS equipped with a ZEISS EVO Series Scanning Electron Microscope EVO 50. HRTEM were performed with PHILIPS CM12 microscope operated at an accelerating voltage of 100 kV. Selected catalysts were subjected to the analysis where TEM images and selected area electron diffraction (SAED) pattern were captured. X-ray diffraction (Rigaku, Miniflex 600) was used to analyze the metal phases of the fresh calcined and spent catalysts at room temperature in the 2θ range from 20° to 80°. CuKα radiation was used as X-ray source generated at 40 kV and 15 mA. Scans were taken with a 2θ step size of 0.02° and a counting time of 1.0 s. The phases of cobalt and iron compounds were identified using database compiled by the Joint Committee on Powder Diffraction Standards (JCPDS). Crystal size of Co_3O_4 and Fe_2O_3 were calculated by the Scherer equation (Eq. (3))

$$d = \frac{0.89\lambda}{\beta\cos\theta} \quad (3)$$

where d is the mean crystallite diameter, λ is the X-ray wavelength (1.54 Å), and β is the full width at half maximum (FWHM) of the diffraction peak [14,15]. Reduction behavior of the catalyst is examined by the temperature programmed reduction (TPR) study, using Micromeritics instrument (Chemisorb 2720, USA) equipped with a thermal conductivity detector (TCD). Same instrument was used for H_2 pulse chemisorption analysis of the samples. Initially 0.1–0.2 g of calcined catalyst was taken in a U tube and degassed at 150 °C for 1 h under flow of argon gas (30 cm^3/min). Then the gas flow switched to H_2 (10.2 vol% H_2 in Argon) with the total flow rate maintained at

30 cm^3/min. The temperature was raised to 800 °C with the ramp rate of 10 °C per minute. The volume of H_2 consumed was plotted against time and temp. Similar method was repeated for H_2 pulse chemisorption analysis. 0.1–0.2 g of degassed sample was reduced in the flow of hydrogen for 3 h at 500 °C estimated from TPR analysis. Then the sample was cooled with argon to 60°C and pulses of pure hydrogen was introduced at equal interval up to saturation (0.1 cm^3). The volume of adsorbed hydrogen used in metal dispersion calculations was calculated using formula given in Eq. (4). XPS spectra were taken by a XPS (SPECS) system equipped with MgKα (1253.6 eV) as the X-ray source. Signal of Si2p line from support was taken as reference signal and was adjusted to 103.4 eV.

TGA (SDT Q600) was used to analyze the properties of spent catalyst. Amount of coke deposited on the surface of the catalyst after 60 h of run was determined using TGA. The major key components present in liquid hydrocarbon synthesized from the Fe-Co catalyst was analyzed by GC–MS (Perkin-Elmer clarus-600) using DB-5 capillary column. The distribution of carbon number in the liquid product were quantified via simulated distillation system using ASTMD2887 method. CHNS Analyzer (Elementar CHNSO Analyzer-vario MACRO cube) was used to determine the carbon content in the wax formed during the reaction.

$$\%\text{metaldispersion} = \frac{H_2 \times \text{atomic weight} \times \text{stoichiometry}}{\%\text{metal}} = \frac{\text{number of metal atoms on surface} \times 100}{\text{total number of metal atoms}} \quad (4)$$

2.3. Catalyst activity test

Fischer–Tropsch synthesis reactions were performed in a fixed bed reactor with 12 mm ID and 29 cm length. The schematic of the reactor set up is shown in Fig. 1. Three individual mass flow controllers (Brooks 5058S Model) were used to control the flow of all three gases (H_2, CO and N_2). Two thermocouples were used for the proper control of temperature inside the reactor. At the exit of the reactor a hot trap system is attached for separating wax formed during the reaction. Next to the hot trap, gas-liquid separator is connected at 1.5 °C temperature. Condensed liquid product is collected from the separator. The entire unit is controlled from a computer using supervisory control and data acquisition system (SCADA) software of General Electric. Flow of the gas at the exit was measured with the help of wet gas totalizer connected at the exit of the whole reactor system. Uncondensed gases at the exit of the reactor were analyzed in gas chromatograph (Nucon 5700) equipped with FID and TCD. The gases H_2, CO, CO_2 and CH_4 were analyzed in GC-TCD after separation in a Carbosieve column, whereas, the lower hydrocarbon C_1-C_5 were analyzed using FID after separation in a Poropack-Q column. The liquid products were analyzed in GC–MS equipped with DB-5 column.

Calcined catalysts were first ground and sieved into a typical size range of 0.5–0.75 mm. 3 g of catalyst was diluted with silicon carbide and loaded in the middle of the reactor with the help of quartz wool and silica beads. The typical height of the catalyst bed was 6–7 cm. The catalyst was reduced in the presence of hydrogen (99.9% pure) at flow rate of 60 mL/min and temperature of 500 °C. The reduction was continued for 6 h at atmospheric pressure. In the next step, temperature and pressure were brought to the reaction condition (220 °C and 20 bar) using inert gas nitrogen. Syngas ratio (H_2/CO) was maintained typically at 1.48. Feed GHSV was maintained at 1200 mL/gcat-h. The reaction was allowed to reach steady state (after 12 h) and then the products were analyzed.

CO conversion and product selectivity (%) were calculated using following Eqs. 5 and 6 respectively. All the calculations were done on carbon basis.

$$\text{COconversion}(\%) = \frac{(\text{moles of CO})_{in} - (\text{moles of CO})_{out}}{(\text{moles of CO})_{in}} \times 100 \quad (5)$$

$$\text{Selectivity of product Ci}(\%) = \frac{(\text{moles of carbon in product Ci})}{(\text{moles of carbon in CO reacted})} \times 100 \quad (6)$$

3. Results and discussion

3.1. Catalyst characterization result

Table 1 summarizes the value of surface area, pore volume and average pore diameter of fresh calcined catalyst. Surface area of monometallic iron and cobalt catalysts are in good agreement with previously reported values in literature where the same method of preparation was used [15,16,18,19]. The value of BET surface area is almost same for the catalyst with 30% metal loading (Cat-1, Cat-2 and Cat-3). Major loss in surface area after addition of metals is observed due to the partial coverage of support by the metal particle. Main dominant phases in the bimetallic Fe-Co catalyst are Co_3O_4 and Fe_2O_3 which were also confirmed through XRD analysis. Presence of other phases, most importantly mixed iron and cobalt catalyst or metal silicates etc. are the main cause of the reduction in surface and these phases are prominent in 15Fe/15Co. Surface area is highest for Cat-7 catalyst, which is attributed to good dispersion and small crystal size of iron oxide and cobalt oxide. Pore diameter of the support material decreases upon incorporation of metal on the support which is result of partial pore blocking with small particles of metal oxide. Also, total pore volume decreases as the Fe/Co ratio reaches 1 (Cat-3) which is attributed to the formation of large metal clusters that block the available pores.

EDS was used to determine the mean elemental composition of prepared catalyst. The elemental compositions of calcined catalysts obtained from the EDS data are shown in Table 2. The EDS analysis of the catalyst confirms the presence of Fe, Co, Si and O. The elemental composition of the metal almost matches the theoretical values.

TEM images of freshly prepared catalyst are shown in Fig. 2. The images of monometallic iron and cobalt catalyst (Cat 4 and Cat 5) were compared with bimetallic Fe-Co catalyst (Cat 2) at same magnifications. The dark spots in the micrograph shows the metal cluster which indicates an uneven distribution of the size of metal cluster. In the bimetallic catalyst, phases of iron and cobalt cannot be distinguished, which shows a proper interaction of the metals. The images were also used for the determination of metal cluster size. The metal cluster size varied from 5 to 15 nm in all the three catalysts which are comparable to the crystal size calculated from XRD (Table 4). SAED pattern of the selected catalyst reveals the good crystalline structure of the catalyst. The concentric rings in the pattern corresponds to atomic planes of different orientation and different inter-planar spacing d. Bright spot at the centre represents the diffracted electrons. The presence of (110) and (211) plane of Fe-Co bimetallic phases are evident in the SAED pattern of Fe-Co bimetallic (Cat-2) catalyst. The magnified electron diffraction patterns was incorporated for the indexing of the SAED pattern. Calculated d-spacing values from the pattern were matched with the standard values obtained from XRD analysis.

TPR profiles of all catalysts are shown in Fig. 3A and B. For monometallic cobalt catalyst Co_3O_4 is the major phase in the oxidized catalysts. It follows the reduction pattern as $Co_3O_4 \rightarrow CoO \rightarrow 3Co$ (Eqs. (7) and (8)). In Fig. 3B TPR profiles d and

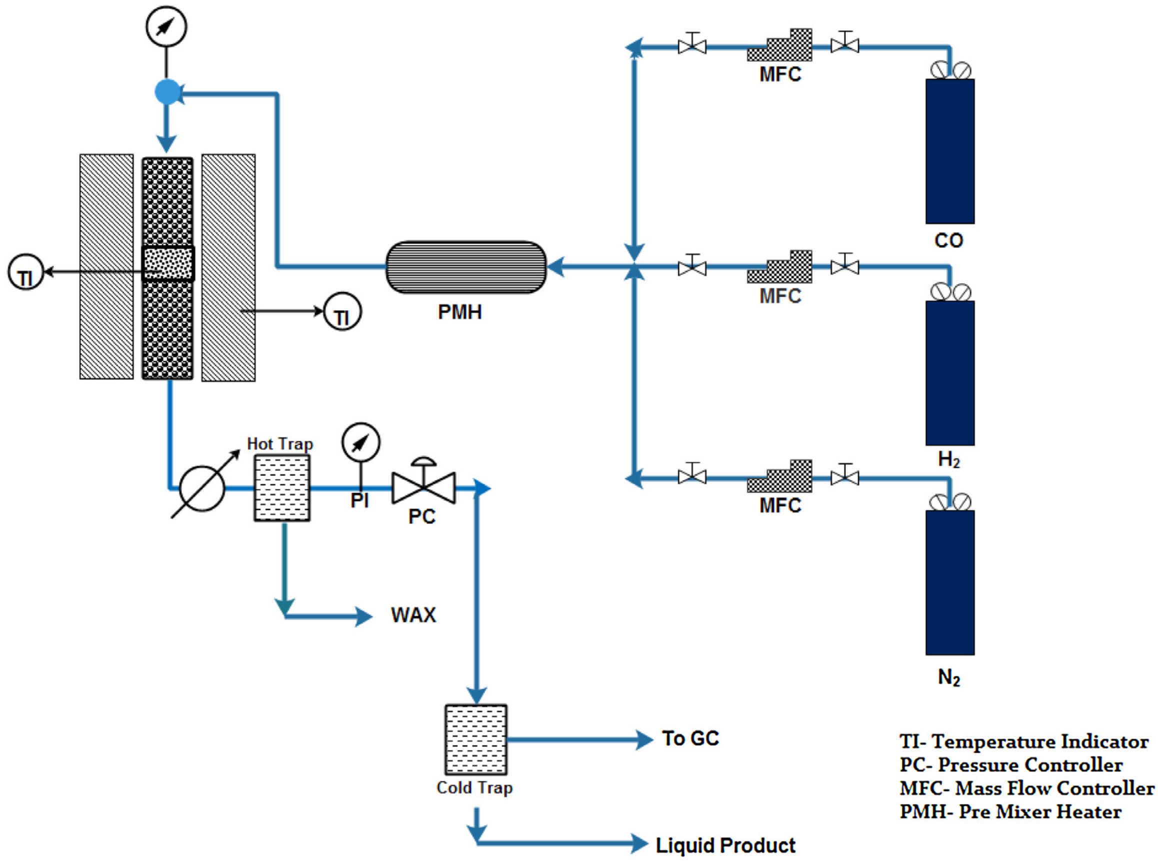

Fig. 1. Schematic representation of fixed bed reactor.

Table 1
BET analysis of freshly calcined catalyst.

Designation	Surface Area (m^2/gm)	Total Pore volume (cm^3/gm)	Av. Pore size (nm)	Metal dispersion, %
Cat-1	193.19	0.40	8.39	1.88
Cat-2	199.72	0.70	14.16	2.43
Cat-3	199.34	0.37	12.17	2.05
Cat-4	211.01	0.78	14.94	1.45
Cat-5	200.82	0.74	14.76	1.56
Cat-6	235.00	0.77	13.16	4.30
Cat-7	245.17	1.04	16.92	4.46

Table 2
EDX analysis of freshly calcined catalyst.

CATALYST		Theoretical wt.%		Observed wt.% (EDX)	
		Fe	Co	Fe	Co
5%Fe-25%Co/SiO_2	Cat-1	5	25	6.0	32.6
10%Fe-20%Co/SiO_2	Cat-2	10	20	11.8	26.8
15%Fe-15%Co/SiO_2	Cat-3	15	15	14.6	17.4
30%Co/SiO_2	Cat-4	–	30	–	30.4
30%Fe/SiO_2	Cat-5	30	–	26.2	–
6%Fe/12%Co/SiO_2	Cat-6	7	14	5.8	12.5
3%Fe/6%Co/SiO_2	Cat-7	3	6	3.0	7.2

Table 3
Crystal size by Scherrer formula.

Catalyst	Co_3O_4 phase			Fe_2O_3 phase			Fe-Co phase	
	d value (Å)	2θ [°]	Crystal size(nm)	d value (Å)	2θ [°]	Crystal size (nm)	d value (Å)	2θ [°]
Cat-1	2.399	37.44	12.08	–	–	–	1.99	45.46
Cat-2	2.413	37.22	12.37	2.50	36.00	1.96	2.00	45.25
Cat-3	2.400	37.44	14.31	2.635	34.00	13.27	1.98	45.57
Cat-4	2.392	37.55	19.40	–	–	–		
Cat-5	–	–	–	2.652	33.76	15.35		
Cat-6	2.423	37.07	5.67	–	–	–		
Cat-7	2.433	36.9	1.33	–	–	–		

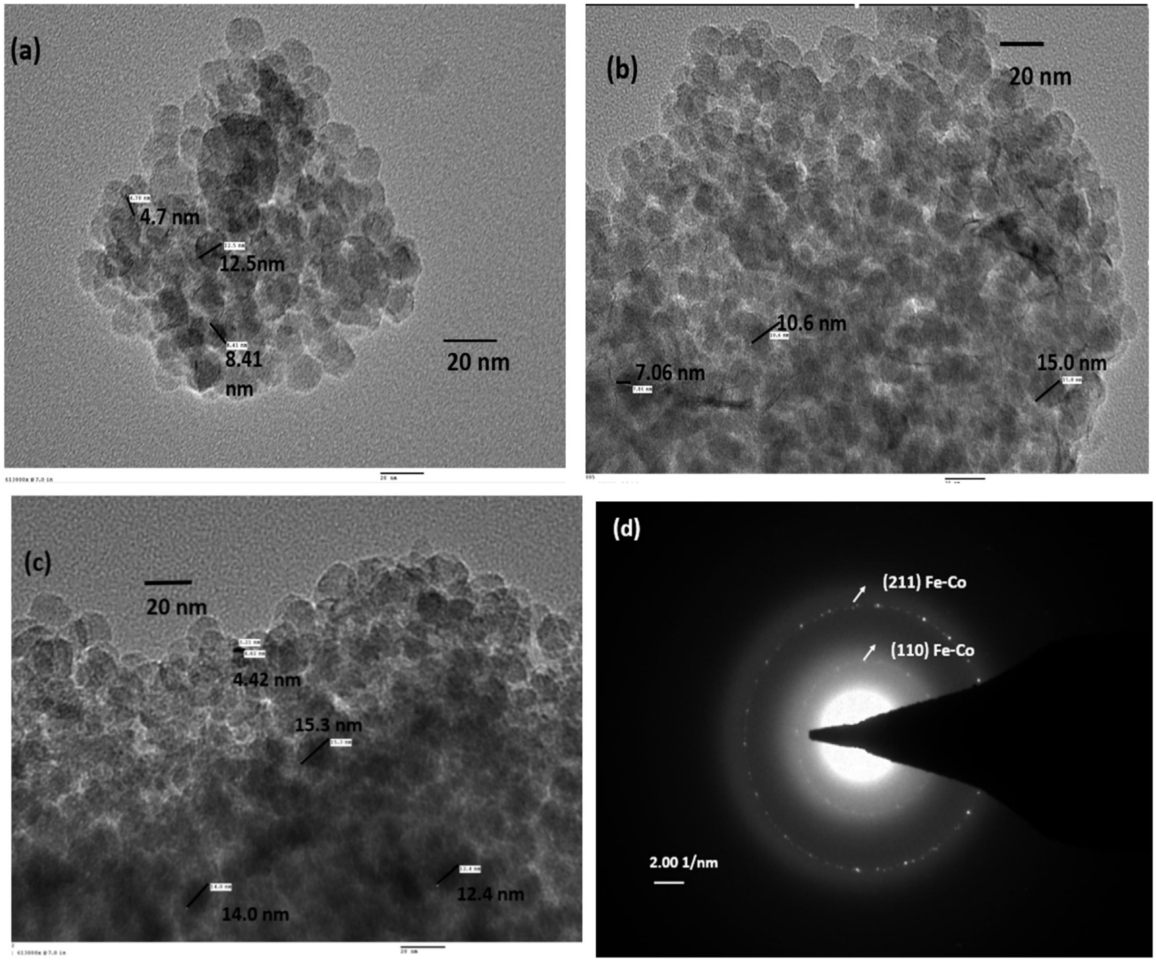

Fig. 2. TEM image of freshly prepared catalyst: (a) $30Co/SiO_2$, (b) $30Fe/SiO_2$ and (c) $10Fe/20Co/SiO_2$, (d) SAED pattern for $10Fe/20Co/SiO_2$.

Table 4
Product Distribution for catalyst with varying Fe to Co ratio (Temp–220 °C, Pressure–20 bar, H_2/CO = 1.48, GHSV = 1200 mL/gcat-h, metal loading –30%).

Catalyst	Cat-1	Cat-2	Cat-3	Cat-4	Cat-5
Fe:Co Ratio	5:25	10:20	15:15	0:30	30:0
%CO conversion	47	65	55.2	45	43.7
%H_2 Conversion	62.3	88.2	71.3	66.5	56.9
% Selectivity					
C_1	8.6	6.4	6	9.6	13
C_{2-4}	7	10	6.5	8.6	4.5
C_{2-4}=	6	4.5	5.8	3.7	10.2
C_{5+} (liq + wax)	68	71	72	75	61
CO_2	8.5	7.1	9.6	2.1	10.9
O/P ratio(C_{2-4})	1.16	2.22	1.12	2.32	0.44
Liquid product(g/min) × 10^{-3}	3.9	6.09	4.8	3.7	3.3
Wax (g/min) × 10^{-3}	1.33	1.53	1.74	1.77	1.03
% C balance	97.74	97.19	96.49	98.35	97.20
H_2/CO Usage ratio	1.96	2	1.92	2.18	1.9

g show the reduction pattern of monometallic cobalt and iron catalyst respectively. Silica supported cobalt catalyst reduces in two steps at 360 °C and 390 °C. Logdberg et al. reported the reduction profile of pure Co_3O_4 and found two peaks at 334 °C and 422 °C, however, addition of silica support shifts the temperature towards higher side due to the metal support interaction. Also, in case of monometallic iron catalyst, reduction of iron oxides takes place according to steps shown in Reactions (8), (9) and (11) in which first Fe_2O_3 was reduced to Fe_3O_4 and Fe_3O_4 to FeO which consequently reduced to Fe. Reduction profile g in Fig. 4B, supported iron catalyst shows major peak at 340 °C and 480 °C and small peak at 660 °C is due to silicate formation. As shown in Fig. 3A and B, the reduction patterns of bimetallic and monometallic catalysts are different. It is difficult to analyze these reduction profiles quantitatively because the iron oxide reduction steps are overlapped by the cobalt oxide ones. In all the three profiles, the two major peaks can be seen which an overlap on the peaks of two reduction steps of iron and cobalt catalyst. Same trend was observed by Ali et al. [22].

The change in reduction temperature of bimetallic catalyst is due to the interaction between metal oxides which formed after alloying the two metal catalyst. It was observed that the cobalt oxide reduction is inhibited by the presence iron oxide because the reduction temperature increases as the iron content increases in the sample. On the contrary reduction of iron is favored by the presence

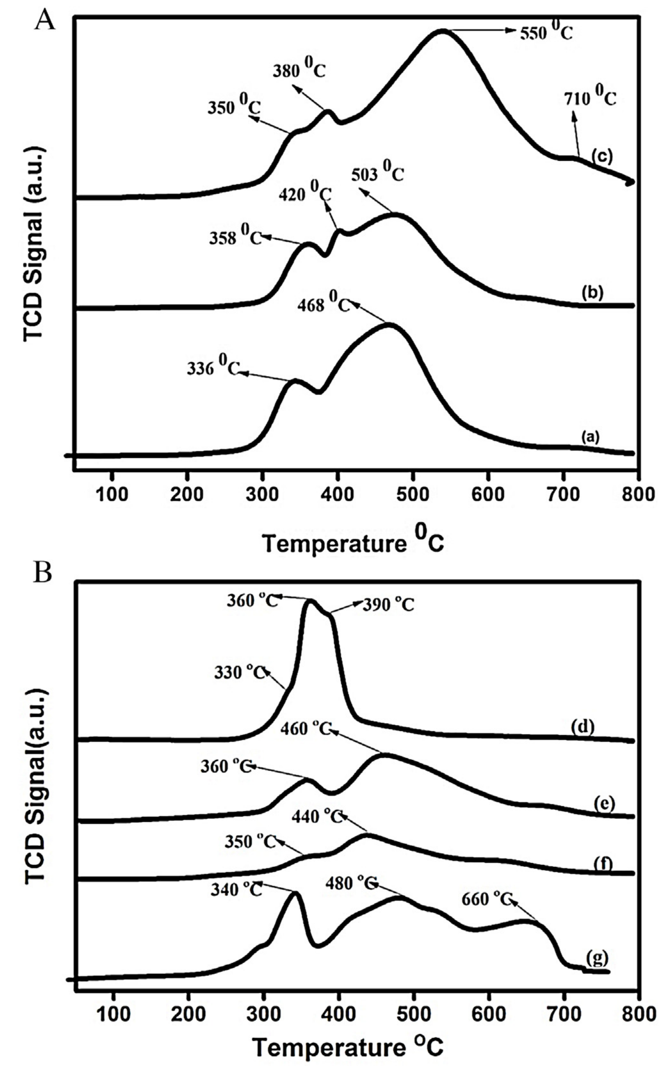

Fig. 3. (A) TPR study of fresh calcined catalyst (a) $5Fe/25Co/SiO_2$ (b) $10Fe/20Co/SiO_2$ (c) $15Fe/15Co/SiO_2$. (B) TPR study of fresh calcined catalyst (d) $30Co/SiO_2$ (e) $6Fe/12Co/SiO_2$. (f) $3Fe/6Co/SiO_2$ (g) $30Fe/SiO_2$.

of cobalt oxide because it is clearly observed that the maximum reduction temperature in all the three profiles is 534 °C which is quite lower than the monometallic iron catalyst.

$$Co_3O_4 + H_2 \rightarrow 3CoO + H_2O \tag{7}$$

$$3CoO + H_2 \rightarrow 3Co + 3H_2O \tag{8}$$

$$3Fe_2O_3 + H_2 \rightarrow 2Fe_3O_4 + H_2O \tag{9}$$

$$3Fe_2O_3 + H_2 \rightarrow 2Fe_3O_4 + H_2O \tag{10}$$

$$6FeO + 6H_2 \rightarrow 6Fe + 6H_2O \tag{11}$$

A small peak was observed at 710 °C in 15Fe/15Co catalyst which could be due the presence of metallic silicates or iron cobalt alloy, i.e. Co_2SiO_4, Fe_2SiO_4 and $CoFe_2O_4$ which reduced at high temperature.

Hydrogen pulse chemisorption was done for all the catalysts in calcined state. It is used to measure the total number of active metal sites exposed on the catalyst surface i.e.% metal dispersion. Table 1 includes the results of the dispersion value of different catalysts. In case of bimetallic catalysts with the metal loading of 30% (Cat-1, Cat-2 and Cat-3), metal dispersion was higher than that of monometallic catalyst (Cat-4 and Cat-5). The results in case of Cat-4 and Cat-5 are very much in agreement with previously reported literature where even less dispersion value was observed [16,20,23]. Metal support interaction also increases with loading. However, addition of iron in the silica based cobalt catalyst increases the total metal dispersion and is maximum for Fe/CO ratio 0.5. The results are also in agreement with XRD results where the crystal size of iron and cobalt oxide in bimetallic system is less as compared with the monometallic system. The decrease in the crystal size directly affects the metal dispersion. A significant change in the dispersion was observed in case of Cat-6 and Cat-7 having metal loading 18%

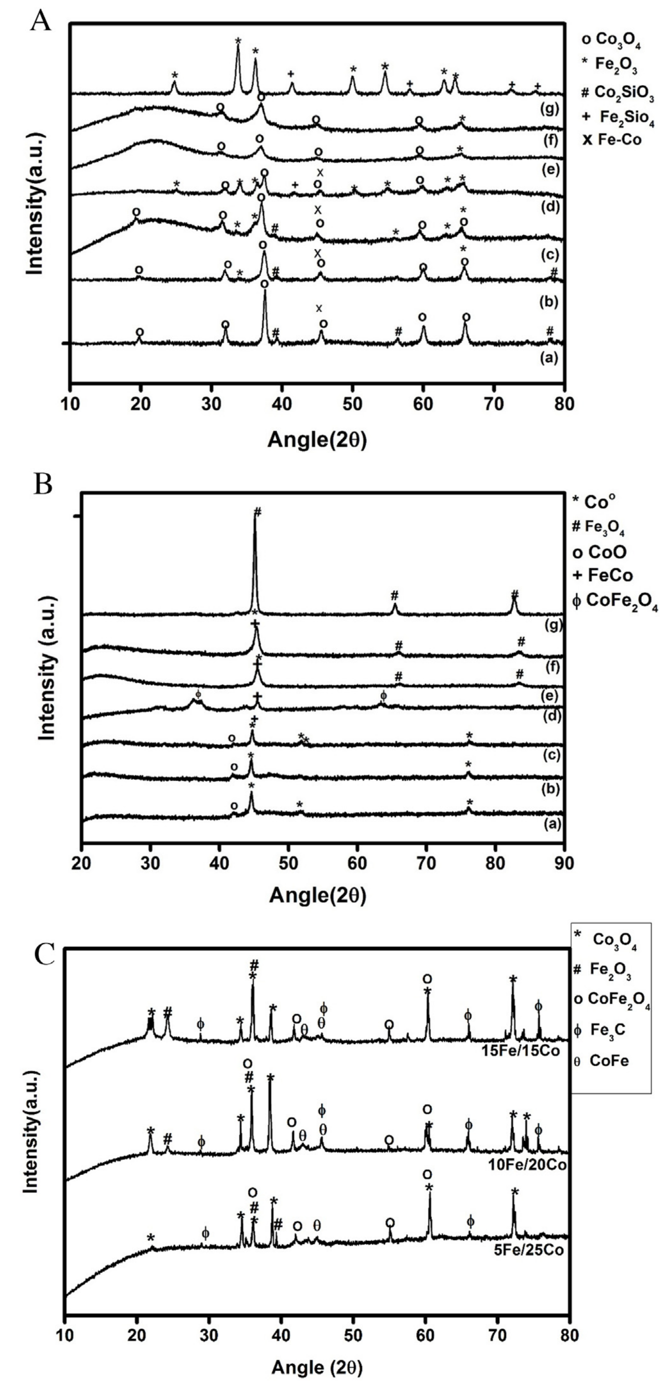

Fig. 4. (A) XRD analysis of fresh calcined catalyst (a) $30Co/SiO_2$ (b) $5Fe/25Co/SiO_2$. (c) $10Fe/20Co/SiO_2$ (d) $15Fe/15Co/SiO_2$ (e) $3Fe/6Co/SiO_2$ (f) $6Fe/12Co/SiO_2$ (g) $30Fe/SiO_2$.(B) XRD analysis of reduced catalyst (a) $30Co/SiO_2$ (b) $5Fe/25Co/SiO_2$. (c) $10Fe/20Co/SiO_2$ (d) $15Fe/15Co/SiO_2$ (e) $3Fe/6Co/SiO_2$ (f) $6Fe/12Co/SiO_2$ (g) $30Fe/SiO_2$. (C) XRD analysis of spent catalyst (a) $5Fe/25Co/SiO_2$ (b) $10Fe/20Co/SiO_2$ (c) $15Fe/15Co/SiO_2$.

and 9% respectively. Decreasing metal loading decreases the crystal size which was confirmed by XRD analysis and the interaction between metal and support also decreases, which ultimately gives rise to increased metallic dispersion.

The analysis of different crystal structures of the silica supported cobalt and iron catalyst were determined by X-ray diffraction. Different phases of metal were evident in the XRD patterns of fresh calcined, reduced and spent catalysts. In Fig. 4A XRD pattern of calcined monometallic cobalt (Cat-4) and iron catalyst (Cat-5) were compared with the bimetallic catalyst having different loading and composition. Strong peaks at 24°, 35°, 39°, 72° and 73° correspond to Fe_2O_3 (JCPDS file no. 85-0599) and peaks at 21°, 36°, 38°, 39.3°, 60°, 73° can be assigned to Co_3O_4 spinel (JCPDS file no.78-1970) in case of monometallic catalyst. It is evident that both the iron and cobalt phases present in the bimetallic catalyst and the intensity of the peak increases as the metal loading increases. In case of Cat-5 and Cat-6, phases of cobalt metal are dominant because, the iron percentage is lower but as the loading increased up to 30% there is a clear picture of both the iron and cobalt phases. Along with the diffraction patterns of Co_3O_4 and Fe_2O_3, there are some lower intensity peaks at 2θ values of 35°, 41°,43°, and 65° which corresponds to formation of $CoFe_4O_4$ bimetallic phases in higher metal loading. Small peak at 45° shows the presence of Fe-Co bimetallic phase which is in agreement of to the observation of Tihay et al. [13], who reported the formation of bimetallic phase without calcination and reduction steps. Effect of composition can also be seen in the XRD pattern. Intensity of the diffraction peaks increases with the amount of iron but decreases as the Fe/Co ratio reaches 1(Cat-3). Overlapping of the peaks (Fe_2O_3 and Co_3O_4) were also observed in case of Cat-3. There may be tendency to form silicates of cobalt and iron, however, the inter planar distance of $CoSiO_4$ and Co_3O_4 are almost same and cannot be differentiated by XRD [24].

In Fig. 4B, XRD profiles of reduced catalysts were compared. All the catalyst were reduced ex-situ in the reactor at 500 °C and taken for the XRD analysis in a close pack chamber filled with inert gas. Diffraction profile of Cat-1(30Co/SiO_2) shows sharp peak at 45° which corresponds to metallic cobalt but the presence of CoO phase proves that the catalyst is not completely reduced. Similar pattern was observed by Khodakov et al. [25] through in-situ reduction of Co/SiO_2 catalyst. A sharp intense peak was present at 45° in case of Cat-7 which corresponds to metallic α-Fe phase. All the bimetallic catalysts exhibited sharp peaks between the 2θ value of 40 and 50 which may correspond to Fe-Co bimetallic phase (JCPDS file no. 44-1433) as there is a strong tendency of formation of this phase after reduction [26]. The presence of metallic iron and cobalt cannot be neglected in the bimetallic catalyst. Hence, the single intense peak in all the bimetallic catalysts can be considered as the result of overlapping of the individual peak as the ionic radii of the most probable phases are very similar [16]. XRD pattern of reduced catalysts provided important information that, the formation of bimetallic phases takes place along with preservation of the individual phases of iron and cobalt active for FT process. Spent catalyst was also analyzed and shown in Fig. 4C for Cat-1, Cat-2 and Cat-3. In the XRD spectra of spent catalyst, it was clearly visible that the catalyst undergoes morphological changes during reduction and also in FT synthesis process. XRD spectra of spent catalyst show many new phases which is due to reduction and long hour of run (60 h) at high pressure. Iron carbide and magnetite (Fe_3O_4) phases can be confirmed in spent catalyst which play pivotal role in FT reaction for hydrocarbon formation and WGS reaction respectively. The diffraction profiles of spent catalyst also displayed diffraction peaks which belong to FeCo mixed phases. In all the three catalysts peaks were observed at 2θ ≈45° corresponding to (110) plane of FeCo crystal with cubic structure (JCPDS file no 49-1567). This indicates that some of the cobalt and iron phases are involved in this mixed phase formation which is the main cause of different behavior of bimetallic catalyst than the monometallic one. These observations are in agreement with the previous studies of Logdberg et al. [3] and Peña O'shea et al. [16].

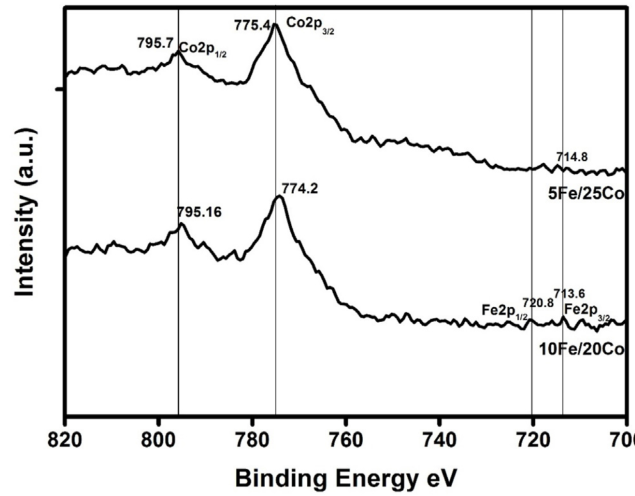

Fig. 5. XPS analysis of fresh calcined catalyst (a) 5Fe/25Co/SiO_2 (b) 10Fe/20Co/SiO_2.

Scherrer equation is used to determine mean particle sizes of cobalt and iron oxides crystals. In case of each oxide, most intense diffraction peak was used for the calculation. The crystal planes taken for calculation are (311) for Co_3O_4 and (222) for Fe_2O_3 with an assumption that the oxide particles have spherical geometry. It was observed that in the case of monometallic catalysts crystal sizes of iron and cobalt oxide are 15 and 19 nm respectively, which decreased after addition of second metal (Table 3). However, in case of 9% and 18% metal loading, crystal size of Co_3O_4 decreased up to 5 nm and the iron phase is below the detection. With the 30% metal loading, crystal size of Co_3O_4 increased up to 12 nm in all the three catalysts while iron oxide was detectable in case of only Cat-3. After reduction and reaction, particle size of the spent catalyst increased due to sintering and agglomeration of particles. The formation of mixed phase, which plays a key role in altering the product distribution, was observed in case of bimetallic catalysts.

Fig. 5 shows the XPS spectra of calcined catalyst. The analyses were done for Cat-1 (5Fe/25Co) and Cat-2 (10Fe/20Co) catalysts to determine the electronic state and relative abundance of metal. The surface enrichment of cobalt metal is clearly evident in the spectra. In the Co_3O_4 spinel Co exhibit Co^{2+} and Co^{3+} electronic state which occupies tetrahedral and octahedral site respectively. In the Fig. 5 two peaks were observed at 775 eV and 795.7 eV due to CO 2p core level spectrum. The separation in the peak is due to spin orbital splitting phenomena which correspond to $Co2p_{3/2}$ and $Co2p_{1/2}$ regions. The intensity ratio and the BE separation value between the peak is dependent on the different electronic state of the metal [27,28]. BE separation value for pure CO_3O_4 compound is 15 eV however, in the present case it was observed to be 20 eV. This shift in the spectra is attributed to metal support interaction. In case of 10 Fe/20Co catalyst, at the 713 eV there is a small peak for $Fe2p_{3/2,}$ however, in case of 5Fe/25Co the peak almost vanished. In both cases, cobalt is the dominant phase. The results are in agreement with XRD analysis where the cobalt phase is dominant and iron phase was below the detection limit for 5Fe/25Co catalyst. XPS analysis is surface sensitive and the depth of the analysis is up to 7–10 nm. This confirms the surface enrichment of cobalt and presence of iron in the bulk in 5Fe/25Co, however, in the case of 10 Fe/20Co catalyst iron is also present on the surface.

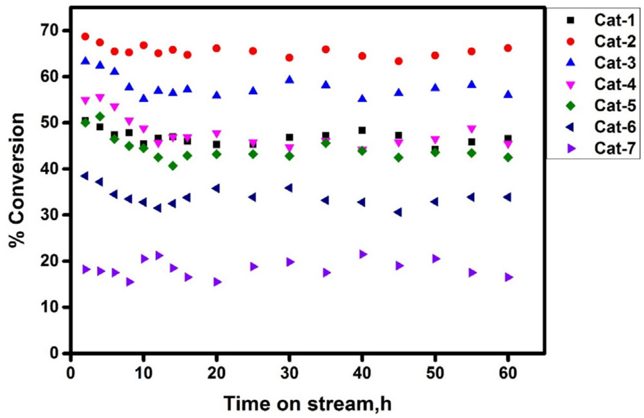

Fig. 6. Conversion vs Time on stream. Reaction conditions: Temp–220 °C, Pressure–20 bar, H_2/CO = 1.48, GHSV = 1200 mL/gcat-h, Run time–60 h.

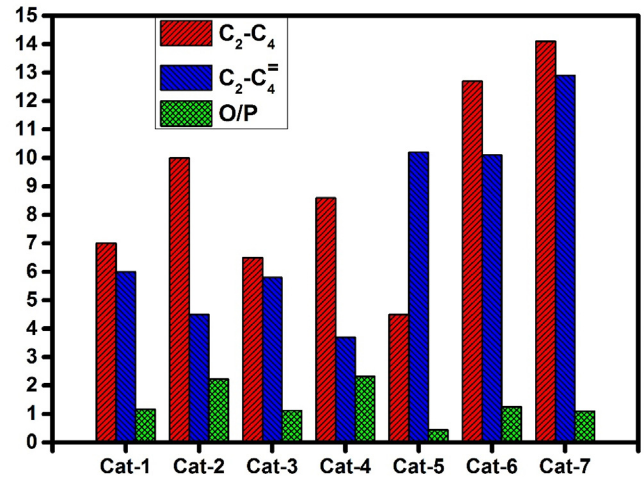

Fig. 7. Paraffin and Olefin distribution. Reaction condition: Temp–220 °C, Pressure–20 bar, H_2/CO = 1.48, GHSV = 1200 mL/gcat-h, Run time–60 h.

3.2. *Catalytic activity testing*

The performances of all the prepared catalysts were investigated in a fixed bed reactor and compared with that of monometallic catalyst. Remarkable improvement in the activity of the catalyst was observed on incorporation of second metal (Fe). Experiments were carried out for the optimization of the Fe/Co ratio and metal loading for maximum C_{5+} selectivity and FT activity. The experimental conditions were maintained as: Temp–220 °C, Press–20 bar, H_2/CO-1.48, GHSV–1200 mL/gcat-h. In the next set of experiments, temperature was varied within the range of 220–250 °C and H_2/CO molar ratio was varied from 0.5 to 2 to achieve maximum WGS activity and C_{5+} selectivity.

Fig. 6 shows the trend of CO conversion with time on stream. In all the cases, negligible deactivation was observed in 60 h of time on stream. However, in case Cat-4 and Cat-5, there is decrease in conversion in initial 5–6 h. The decrease in conversion can be explained on the basis of pore filling phenomena. In the initial hours of the run, the catalyst remains highly active and as the pore fills with the liquid product, activity decreases. In spite of the filled pores, catalysts remains active, however, H_2 and CO have to diffuse through the liquid to reach the active site. This period of pore filling depends on pore size, particle size and extent of mass transfer diffusion resistance. Madon et al. [29] observed that for small pellets, (0.11–0.18 mm) pore filling time was less than 2.5 h, while, for large pellets (0.85–1.7 mm), the time was more than 20 h because the later one was under the influence of strong mass transfer diffusion limitation. In the present study, the catalyst particle size (0.2–0.5 mm) falls in the region of negligible diffusion resistance [26,27,30]. In case of bimetallic catalysts, initial drop in CO conversion was less; however, the mass transfer diffusion resistance is not negligible. In case of Cat-7, initial decrease in CO conversion was observed. This may be due to the increased pore size of Cat-7 after metal impregnation. In all the cases, steady state conversion was taken for the further calculation.

3.2.1. *Effect of metal composition*

Table 4 includes the product distribution and CO conversion for the catalysts. Bimetallic catalysts Cat-1, Cat-2 and Cat-3 were compared with monometallic catalyst Cat-4 and Cat-5 at similar metal loading (30%). The conversion was calculated using equation 1 for the period of 60 h and product selectivity was calculated using equation 4. In all the cases carbon balance was about 95%. It was observed that the CO conversion was maximum (65%) over the catalyst with Fe/Co ratio 0.5 (Cat-2), but increasing the Fe content at constant metal loading (30%) shows negative effect on CO conversion. In case of Cat-4 and Cat-5 the conversions were 45% and 43% respectively, which is in agreement with literature related to monometallic iron and cobalt catalysts. Sun et al. [31] examined the effect of cobalt loading on Co/SiO_2 catalyst and observed that conversion increased from 1.38 to 56.84% when Co loading was increased from 2 to 40 wt.% and for 20% Co loading conversion was 29.83%. While showing the effect of alkali promoters, Das et al. [20,21] showed that without any alkali promotion upon silica based iron catalyst, only 38% CO conversion along with 48% C_{5+} selectivity was possible (T–245 ± 3 °C, H_2/CO-2, P–15 bar). Methane selectivity was maximum in case of monometallic catalysts (Cat-4 and Cat-5), however, in case of bimetallic catalysts, it decreased significantly. Selectivity to C_{5+} hydrocarbon is also shown in Table 4 which includes the moles of carbon in liquid and wax product. Selectivity to C_{5+} hydrocarbon over bimetallic catalysts were between 68–71% and was maximum for Cat-2. There was no remarkable change in amount of wax when Cat-2 was compared with Cat-4, however, rate of formation of liquid product (C_5-C_{20}) increased significantly. Cat-2 and Cat-4 have almost comparable C_{5+} selectivity, however, there is significant increase in the amount of liquid product. Cat-5 has lowest C_{5+} selectivity with significant amount of wax. Olefin and paraffin distribution for C_2-C_4 gaseous product is shown in Fig. 7. The probability of long chain formation and termination depends on the initial compounds. In the bimetallic catalysts, selectivity to paraffin (C_2-C_4) was maximum (10%) for Cat-2 while the olefin percentage was lowest for the same. In case of monometallic catalysts, paraffin selectivity was high for Cat −4 while, olefin selectivity was significantly high in case of Cat-5. From above discussion it can be concluded that performance of Cat-1 is inclined towards Cat-4 as the cobalt content is high in the catalyst. It is also confirmed through XRD analysis that the cobalt phases are dominating in Cat-1. However, as the iron content increases (Cat-2 and Cat-3), the impact of bimetallic phase formation is evident in the results.

The trends in product selectivities and CO conversion over bimetallic Fe-Co catalyst can be explained by various factors: (i) intrinsic rate of water gas shift (WGS) reaction changes the H_2 to CO ratio on the catalyst surface (ii) modification of some surface Co and Fe sites by the presence the other metal in the catalyst. (iii) Change in structural properties such as metal dispersion, pore size and crystal size.

When compared with monometallic catalyst, CO conversion was observed to be high in case of bimetallic catalysts which can be related to the increased metal dispersion (Table 2). Metal dispersion is maximum (2.43%) at Fe to Co ratio 0.5 when compared to constant metal loading. It decreases as iron content increases. Methane is an undesirable product in the FT synthesis but is highly favored thermodynamically. Methane production significantly decreased over these bimetallic catalysts. Formation of methane depends on the H_2 to CO ratio in the vicinity of the active sites and weak CO adsorption. In case of Cat-5 high CH_4 selectivity was due to high WGS activity of iron catalyst which increases the local concentration of H_2 [32]. Over the bimetallic catalysts CO adsorption is relatively strong due to iron cobalt interaction which suppresses the methane production. While investigating the CO adsorption theoretically on Fe-Co bimetallic surface, Rochana et al. [33] used density functional theory to prove that the presence of iron around the cobalt facilitates the CO adsorption on cobalt surface. CO adsorption energy also increases in case of bimetallic Fe-Co phase in comparison to only iron or cobalt phases due to which probability of formation of long chain hydrocarbon increases and amount of methane decreases. Experimentally several authors have observed the same trend [2,7,26]. Arai et al. [7] and Ishihara et al. [2] observed increased C_{5+} selectivity in case of bimetallic catalyst. At the same time some authors observed opposite trends in selectivity and observed that it shifts towards lower hydrocarbon [3,34,35].

The olefin to paraffin ratio (O/P) is an important parameter, on which the anticipation of liquid product distribution depends up to a great extent. Recent studies in the FT process suggested that the FT mechanism divided in the three parts, CO activation, chain propagation and chain termination. Chain termination can either takes place via β-hydride elimination which ends up forming α-olefin or via α-hydrogen addition for the release of paraffin. There is high probability of the readsorption of the α-olefins which either terminate as paraffin or initiate new growing chain for long chain hydrocarbon. So the quantitative measurement of olefin, paraffin and the O/P ratio in gas phase is required for the estimation of liquid product distribution [36,37]. Fig. 7 shows the distribution of gaseous product in range of C_2-C_4 hydrocarbon. When compared Cat-1 to Cat-5, maximum saturated hydrocarbon was produced over Cat-2 while olefin content was almost similar for Cat-1 Cat-2 and Cat-3.O/P ratio was maximum (2) for Cat-2 which indicates that the olefin readsorption and their subsequent hydrogenation into paraffin is taking place frequently on bimetallic catalyst. It is also reflecting in liquid product stream as the selectivity to C_{5+} fraction is maximum for Cat-2. Formation of lower range olefin, alkane and hydrogenation of reabsorbed olefin depends on the H_2/CO ratio in the vicinity of catalyst active site [38]. Presence of considerable amount of CO_2 in the product stream, confirms that WGS reaction is taking place inside the reactor [39]. Due to WGS reaction inside the reactor, H_2/CO ratio increases at the active site. It increases the hydrogenation of olefin into paraffin. Higher C_2-C_4 alkane selectivity in case of Cat-1 is due to lower iron content in the catalyst and all the iron involved in WGS activity. Monometallic iron catalyst shows highest selectivity C_2-C_4 olefin and paraffin due to very high WGS activity.

3.2.2. *Effect of Fe/Co loading*

Catalysts with varying metal loading (Cat-2, Cat-6 and Cat-7) at similar Fe to Co ratio (0.5) were compared in the Table 5. In case of lower metal loading instead of high metal dispersion the% CO conversion was very less. CO conversion over Cat-6 and Cat-7 showed a sharp decrease when metal loading was decreased to 18% and 9% respectively at constant Fe to Co ratio. This can be attributed to the crystal size and extent of reduction. In case of small crystal size metal-support interaction is very high which plays negative role in the reducibility [40,41]. Also, very small cobalt crystal is prone to deactivation by oxidizing through the water molecule [41]. It is evident in Table 3 that the crystal size of cobalt oxide is very small in case of Cat-6 and Cat-7 while size of iron oxide is not detectable. It is important to observe that lowering the metal shifts the product distribution toward lower range hydrocarbon (C_1-C_4). C_{5+} selectivity remain between 56 and 60%. Bimetallic system does not help to bring down the methane level in the product stream when metal loading is low. Selectivity to gaseous paraffin is almost comparable over Cat-6 and Cat-7. Rate of formation wax decreased remarkably in case of Cat-6 and Cat-7 at the expense of lower rate of liquid product formation. Iron shows good WGS activity with the significant amount of CO_2 in product range. However, low CO conversion over Cat-6 and Cat-7 made the catalysts inappropriate for further consideration.

Table 5
Product Distribution for catalyst with varying metal loading (Temp–220 °C, Pressure–20 bar, H_2/CO = 1.48, GHSV = 1200 mL/gcat-h, Fe/Co ratio-0.5).

Catalyst	Cat-2	Cat-6	Cat-7
% Metal Loading	30	18	9
%CO conversion	65	35.9	20.6
%H_2 Conversion	88.2	52.6	29.7
% Selectivity			
C_1	6.4	12.6	11.4
C_{2-4}	10	12.7	14.1
C_{2-4}=	4.5	10.1	12.9
C_{5+}	71	60	56
CO_2	7.1	3.45	3.5
O/P ratio(C_{2-4})	2.2	1.25	1.09
Liquid product(g/min)x10^{-3}	6.09	2.7	1.38
Wax (g/min) x10^{-3}	1.53	0.84	0.52
% C balance	97.19	96.08	97.27
H_2/CO Usage ratio	2.0	2.16	2.14

3.2.3. *WGS activity over Fe-Co bimetallic catalyst*

Addition of iron in cobalt catalyst increases WGS activity inside the reactor. In order to measure the WGS activity of catalyst, H_2/CO usage ratio the exit of rector can be used. It is the ratio of moles of H_2 consumed per mole of CO consumed [42]. Considering Eqs. (1) and (2), there are two extreme value of the H_2/CO usage ratio, 0.5 and 2. When there is no WGS reaction taking place, theoretically the ratio would be 2 and when all the water forming in reaction 1 is consumed through Reaction (2) then the ratio would be 0.5. At 220 °C WGS activity is very less in case of bimetallic catalysts. As shown in Table 4, % CO_2 selectivity is between 7 and 10% and usage ratio is between 1.9–2.0. In case of monometallic iron catalyst (Cat-5) the CO_2 formation is significant which confirm the occurrence of WGS activity. On the contrary in case Cat-4, CO_2 formation is negligible. In the bimetallic catalysts, % CO_2 selectivity increases with the amount of iron. The presence of active phase (Fe_3O_4) for WGS reaction is clearly evident in the XRD analysis of catalysts. Hence the WGS activity increases with the percentage of iron content and was maximum for Cat-3. However, in the term of C_{5+} selectivity and WGS activity, Cat-2 was found to be most appropriate catalyst. In case of Cat-2, H_2/CO usage ratio was 2.0 and the C_{5+} selectivity was 71% (Table 4).

Rate of WGS rection is highly dependent on temperature. To investigate the effect of temperature on WGS activity and C_{5+} selectivity of the Cat-2(10Fe/20Co/SiO_2), temperature was varied from 220 to 280 °C at Pressure–20 bar, GHSV 1200 mL/gcat-h and H_2/CO ratio of 1.48. The effect of temperature on% CO conversion, methane selectivity and% C_{5+} selectivity is shown in Fig. 8. The variation in selectivity to CO_2 and H_2/CO usage ratio with the temperature is shown in Fig. 9.% CO conversion, increases up to 260 °C and was almost constant for 280 °C. At the same time methane selectivity increases whereas C_{5+} selectivity decreases with the temperature. In order to select appropriate temperature at which C_{5+} selectivity remain maximum along with minimum production of methane, the effect of WGS reaction is needed to be considered. WGS activity increases significantly from 220 °C to 240 °C with the increased selectivity to CO_2 (7% to 20%). However, after 240 °C the increase in WGS activity was comparatively slow and the H_2/CO usage ratio

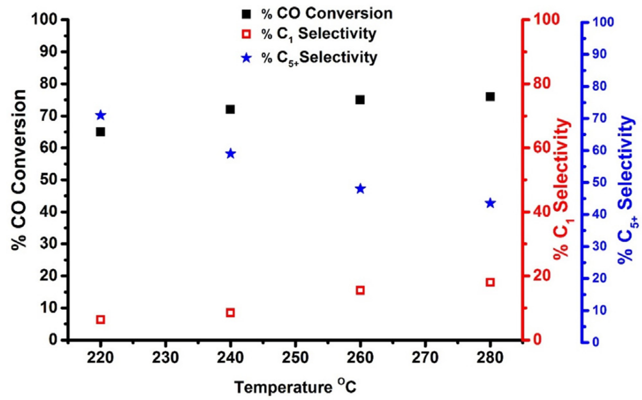

Fig. 8. % CO Conversion, % C_1 Selectivity and% C_{5+} Selectivity vs Temperature at Pressure–20 bar,H_2/CO = 1.48, GHSV = 1200 mL/gcat-h, catalyst- 10Fe/20Co/SiO_2.

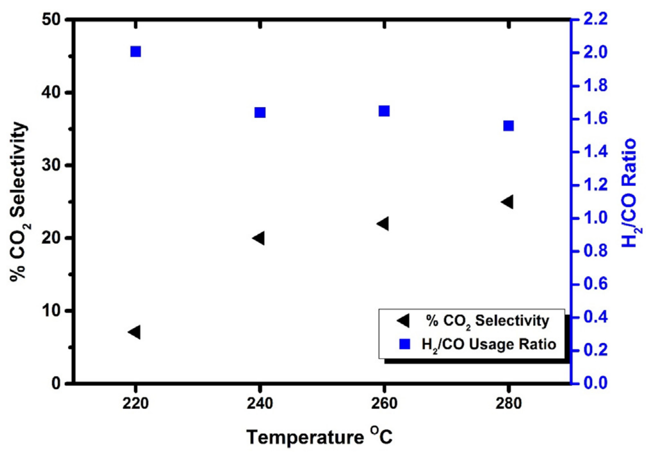

Fig. 9. % CO_2 Selectivity and H_2/CO Usage ratio vs Temperature at Pressure–20 bar, H_2/CO = 1.48, GHSV = 1200 mL/gcat-h, catalyst-10Fe/20Co/SiO_2.

was almost 1.6. So the increased amount of H_2/CO ratio was comparable for the last three temperatures. In order to select optimum temperature, selectivity to C_{5+}, C_1 and CO_2 were considered. At 240 °C, 72% of CO converted into C_{5+} hydrocarbon and selectivity to methane and CO_2 were 8% and 20% respectively. So the increased H_2 concentration due to water gas shift reaction is utilizing in formation of long chain hydrocarbon.

Since the work is to utilize poor H_2/CO ratio feed stock, therefore it is required to investigate the effect of range of H_2/CO ratio at the inlet. The molar ratio was varied within range of 0.5–2.0 at identical reaction conditions (T–240 °C, P–20 bar, GHSV- 1200 mL/gcat-h). Our results show that (Fig. 11) the increased H_2/CO ratio increases formation of lower hydrocarbon. However C_{5+} formation does not increased significantly. Liu et al. [43] suggested that the increased H_2/CO ratio facilitates the chain termination over propagation leads to maximum lower chain hydrocarbon (C_1-C_4).% CO conversion increased significantly from 28% to 72% then decreases to 70%. The reason is that, at lower H_2/CO ratio (0.5) hydrogen is limiting reactant and control the CO conversion while at H_2/CO = 2, CO is limiting reactant. Fig. 10 show the effect of inlet feed composition on WGS activity. Increased selectivity to CO_2 and decreased H_2/CO usage ratio indicate that the activity WGS reaction is high when the H_2/CO ratio is between 1 and 1.48. At the same time selectivity to C_{5+} fraction is almost similar when the inlet feed ratio varied from 1 to 2. So the addition of H_2 via WGS reaction is evident in case of low inlet feed ratio H_2/CO ratio (1 and 1.48). However, maximum CO conversion (72%) at H_2/CO ratio of 1.48 made this feed composition optimum for the process.

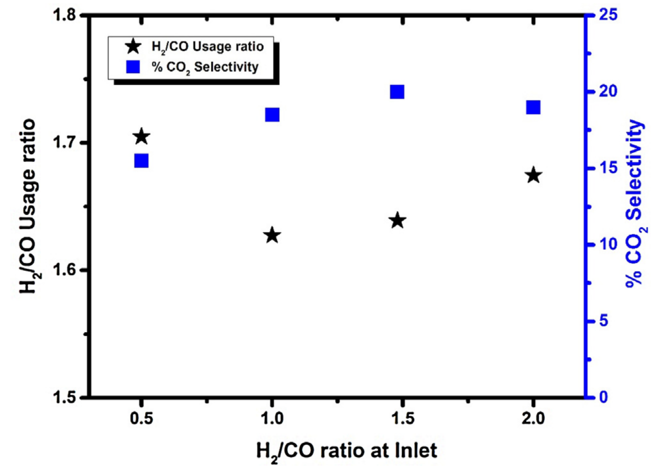

Fig. 10. % CO_2 Selectivity and H_2/CO Usage ratio vs H_2/CO ratio at Pressure–20 bar, Temp–240 °C, GHSV = 1200 mL/gcat-h, catalyst-10Fe/20Co/SiO_2.

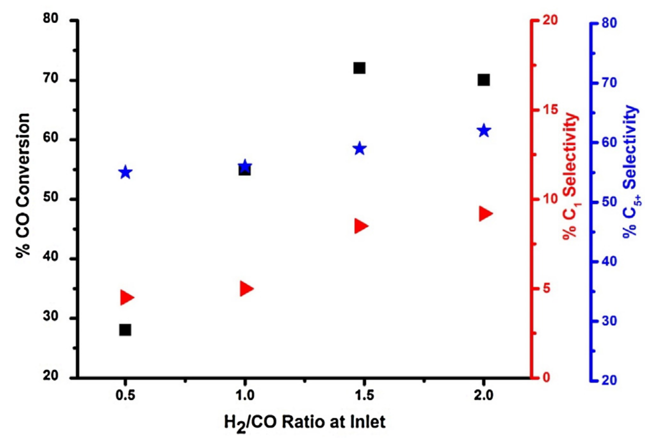

Fig. 11. % C_{5+} and C_1 Selectivity and% CO Conversions vs H_2/CO ratio at Pressure–20 bar, Temp–240 °C, GHSV = 1200 mL/gcat-h, catalyst-10Fe/20Co/SiO_2.

In the view of above discussion it is observed that the Cat-2 (10Fe/20Co/SiO_2) with 30% metal loading and Fe to Co ratio of 0.5, is most suitable catalyst for the production of maximum C_{5+} selectivity and CO conversion. At the temperature of 240 °C and H_2/CO usage ratio, Cat-2 shows maximum WGS activity which leads to increased H_2/CO intrinsically.

3.3. *Characterization of spent catalysts and product*

3.3.1. *Thermogravimetric analysis of spent catalyst (TGA)*

Thermal gravimetric analysis (TGA) was used to investigate the thermal stability of the catalyst. TGA curve for spent catalysts are shown in Fig. 12. During the long hour run (~60 h) there was some deposition of carbon on the active sites which slowed down the catalyst activity. Prior to the analysis catalyst were pretreated with CS_2 in order to remove wax accumulated in the pores of catalyst. Catalysts were dissolve in CS_2 and ultra-sonication were done for an hour in case of each sample.

Fig. 12 is the TGA curve of spent catalyst which clearly indicates the difference in the amount of carbon deposited. The weight loss is lowest (10%) for the Cat-6 (6Fe/12Co) and Cat-7 (3Fe/6Co). Weight loss is between 18–20% in case of Cat-1, Cat-2, and Cat-3. Major weight loss (30%) was observed in case of monometallic iron and cobalt catalyst (Cat-4 and Cat-5).

Table 6
Liquid hydrocarbon products by GCMS analysis.

RT (min)	Molecular weight (approx.)	Formulae	Possible compounds
7.15	142	$C_{10}H_{22}$	Decane
10.67	156	$C_{11}H_{24}$	Undecane
12.95	170	$C_{12}H_{26}$	Dodecane
13.93	184	$C_{13}H_{28}$	Tridecane
14.71	198	$C_{14}H3_{0}$	Tetradecane
16.22	212	$C_{15}H_{32}$	Pentadecane
17.56	226	$C_{16}H_{34}$	Hexadecane
18.79	240	$C_{17}H_{36}$	Heptadecane
19. 95	254	$C_{18}H_{38}$	Octadecane
21.04	268	$C_{19}H_{40}$	Nonadecane
22.07	282	$C_{20}H_{42}$	Eicosane
23.05	296	$C_{21}H_{44}$	Eicosane-8-methyl
24.00	296	$C_{21}H_{44}$	Eicosane-10-methyl
24.90	296	$C_{21}H_{44}$	Henicosane
25.84	296	$C_{21}H_{44}$	Hepatadecane,2,6,10,15-tetramethyl-
26.97	338	$C_{24}H_{50}$	Tetracosane
28.36	352	$C_{25}H_{52}$	Hepatadecane-9-octyl-
30.12	478	$C_{34}H_{70}$	Tetracosane-11-decyle

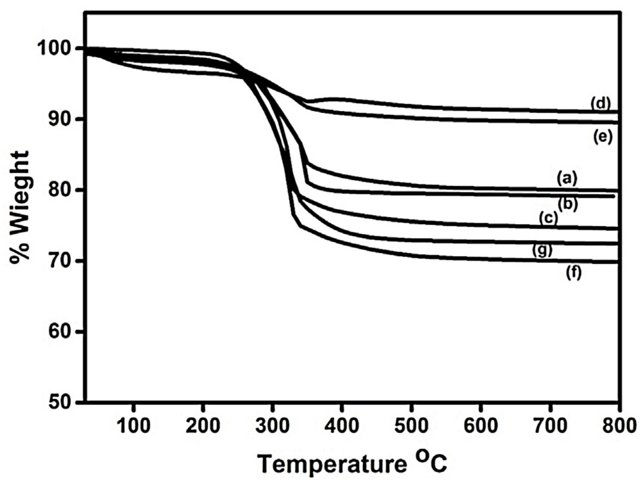

Fig. 12. TGA analysis of spent catalysts (a) $5Fe/25Co/SiO_2$ (b) $10Fe/20Co/SiO_2$ (c) $15Fe/15Co/SiO_2$ (d) $3Fe/6Co/SiO_2$ (e) $6Fe/12Co/SiO_2$ (f) $30Co/SiO_2$ (g) $30Fe/SiO_2$.

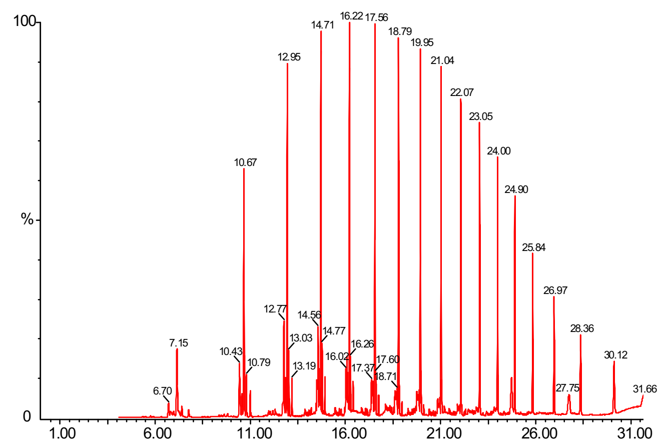

Fig. 13. Liquid hydrocarbon identification by GCMS analysis obtained from 10Fe/20Co (Cat-2) catalyst.

3.3.2. Liquid product analysis

Qualitative analysis of liquid products was done by GC–MS (Model: Perkin–Elmer clarus-600) analyzer equipped with DB-5 column. Liquid hydrocarbon product obtained from all the catalysts was collected at end of each run (about 60 h) for analysis. CS_2 was used as the solvent for the dilution of the sample. The typical chromatogram for Cat-2 is shown in Fig. 13 and the most probable compound given by the library at different retention times are given in Table 6. Major peaks of the chromatograms correspond to linear chain paraffins C_{10}-C_{21}. Each intense peak surrounded by small peaks which correspond to higher olefin.

Simulated distillation analysis of selected product were performed for the quantitative analysis of liquid product. ASTMD2887 method was employed for the analysis. Fig. 14 shows the distribution of the carbon number in the liquid product of the Cat-1, Cat-2 and Cat-3. The results reveal that the distribution is similar in all the three case, however, the mass fraction of carbon number C_{11}-C_{19} is maximum (75%) in case of Cat-2.

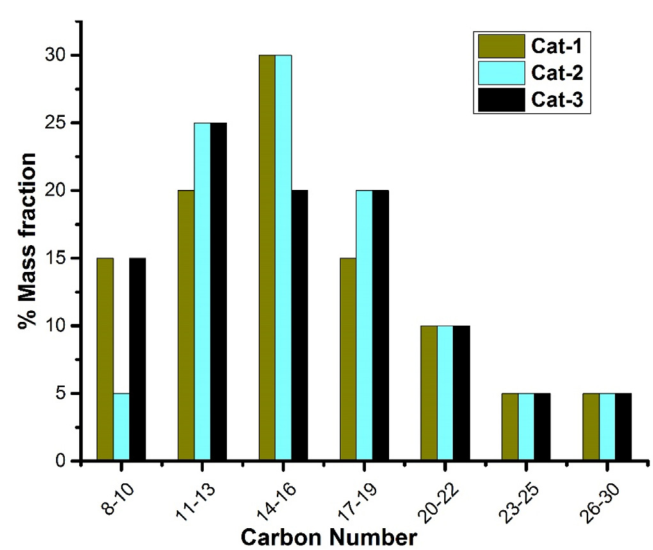

Fig. 14. Distribution of liquid product for Cat-1, Cat-2 and Cat-3.

3.3.3. CHNS analysis of wax product

Considerable amount of wax was produced in case of all the three catalysts at the end of the run. In order to quantify the amount of carbon and hydrogen in the wax, CHNS analysis has been performed. Solid wax was kept in oven at 100 °C temperature for half an hour in order to remove traces of water. A known quantity of each sample was analyzed thrice to check the reproducibility of the results. Table 7 shows the results of the analysis in weight% of carbon and hydrogen. The presence of nitrogen and sulphur were negligible. The percentage of carbon in all the three samples ranges

Table 7
CHNS analysis of wax product.

Catalyst	Amount of wax (g)	Carbon (wt%)	Hydrogen (wt%)	Sulphur (wt%)	Nitrogen (wt%)
Cat-1	7.5	84.01	14.9	–	0.09
Cat-2	9.6	83.08	15.25	–	0.05
Cat-3	9.1	84.36	13.8	–	0.12
Cat-4	10.2	83.45	16.4	–	0.45
Cat-5	7.6	82.56	17.2	–	0.04
Cat-6	5.9	85.6	13.8	–	0.06
Cat-7	3.1	84.6	14.5	–	0.03

between 84 and 85% and that of hydrogen is between 13 and 14% and the rest being impurities. It was observed that Fe/Co ratio in the catalyst does not really affect the nature of wax. The data was used for the confirmation of material balance.

4. Conclusion

Fe-Co bimetallic catalyst supported on silica showed a high activity when compared to the monometallic catalysts. The advantage of addition of Fe into Co catalyst is seen through increased WGS activity which increases H_2/CO ratio and shifts the product range towards C_{5+} range hydrocarbon. Optimum ratio of iron to cobalt in silica supported catalyst was found to be 0.5 in 30% total metal loaded catalyst. At 220 °C maximum CO conversion and C_{5+} selectivity obtained was 65% and 72% respectively over Cat-2. When the metal composition reaches the ratio 1 (Fe/Co) there was a decrease in the performance suggesting that after a certain level of iron addition, the mutual interaction of metals strengthens which cause a negative effect. Presence of cobalt facilitates the reduction of iron however, presence of iron enhances the reduction temperature of cobalt. When the catalyst loading is low (9%), the iron and cobalt dispersion is high due to separate phases of the metals. Presence of iron does not affect the performance of cobalt and vice versa. CO conversion decreases with metal loading which is due to small metal crystal size which easily re-oxidizes in presence of water. With increasing metal loading (Cat-6 and Cat-2) interaction between the metal increases which end up forming Fe-Co bimetallic phase and the formation of long chain hydrocarbon facilitate. High CO_2 selectivity over Cat-6 and Cat-7 shows that the iron catalyst is active toward WGS reaction even when the percentage is low. However, the% CO conversion was low in these case. At 220 °C, activity of bimetallic catalyst for WGS reaction was low. With the increased temperature activity towards WGS rection increased significantly. At 240 °C temperature and H_2/CO of 1.48 Cat-2 showed maximum CO conversion (72%) with maximum C_{5+} (60%) selectivity.

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